NxM Templates for SuperCDMS SNOLAB Run 4 — Ge Activation Data

K-line event selection · free-pretrigger two-exponential fits · 1x1 and NxM (PCA) templates

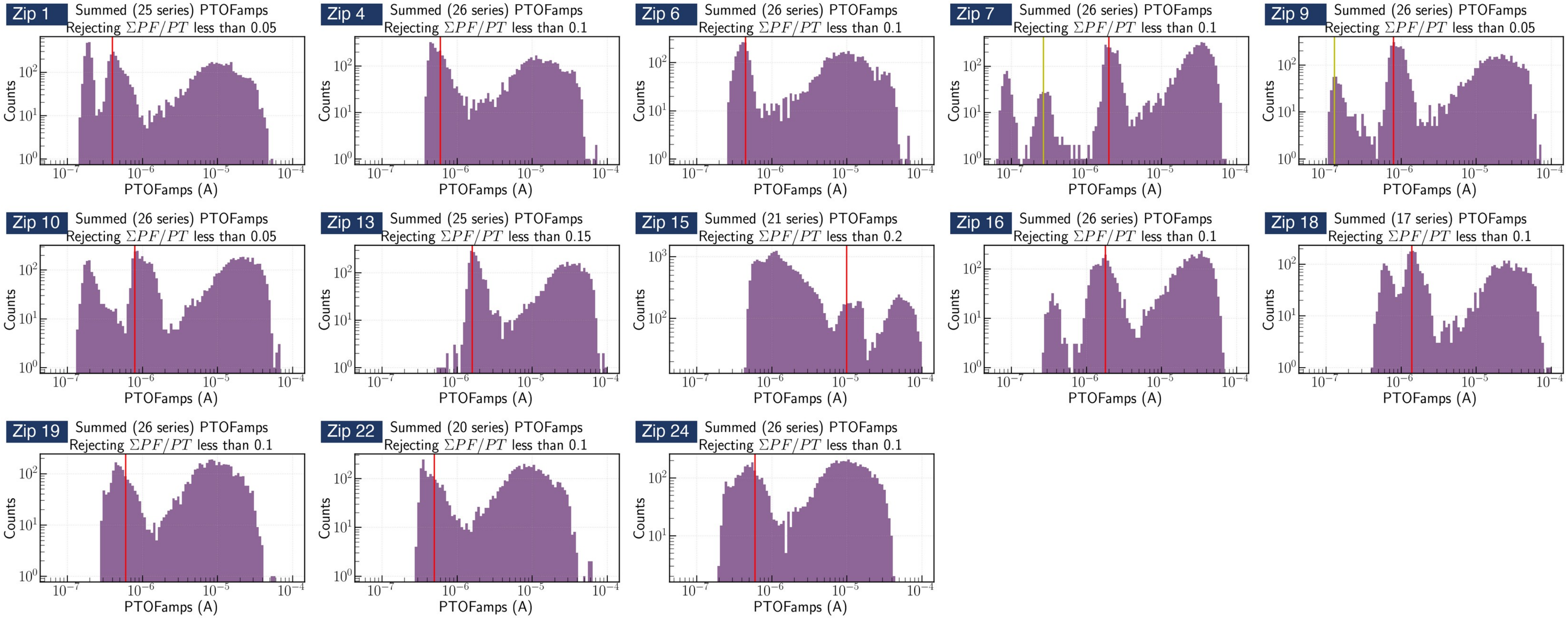
Zhiheng Li — July 2026

From the Ge-activation K-line to an event sample

Data: Ge Activation Data — Ops Shift 260612
[Credit: Prof. Tarek Saab · SuperCDMS Confluence]

Goal: one clean, minimally-biased pulse population per detector × channel

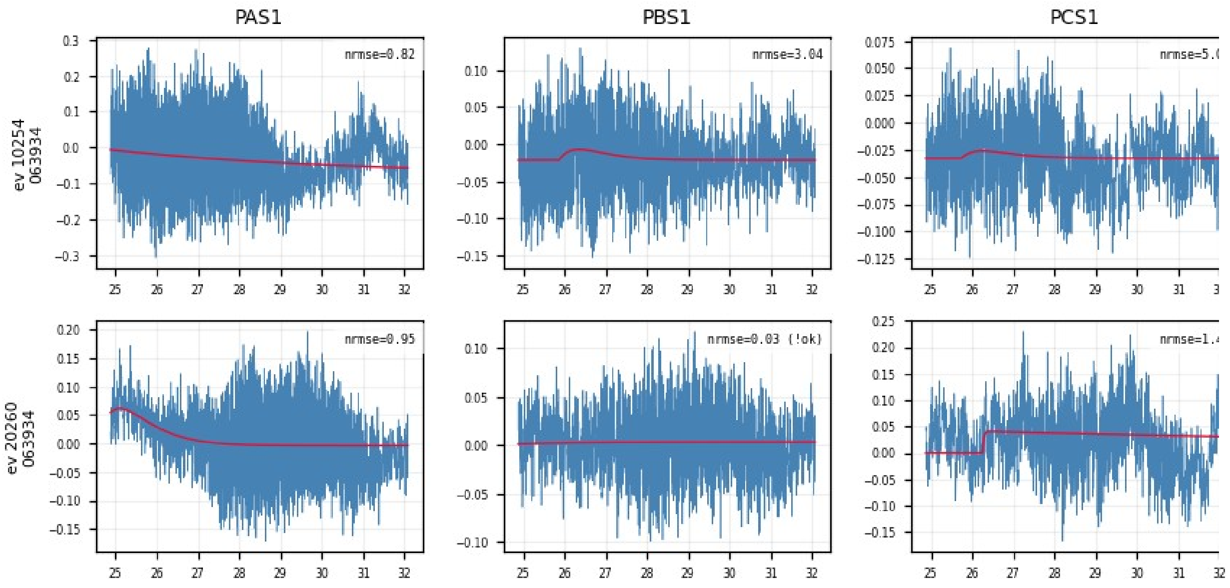
- Per detector, Prof. Saab's study marked the 10.37 keV K-line position in PTOFamps — the red line in each panel below
- Our event selection: a PTOFamps window bracketing the red line — position ×/÷ 1.35, a rough, somewhat arbitrary eyeballed choice; deliberately minimal, no other cuts
- For every selected event: cache the fully unprocessed raw MIDAS traces of all channels + event metadata
- 13 detectors (zips) × 27–30 series — full raw traces cached



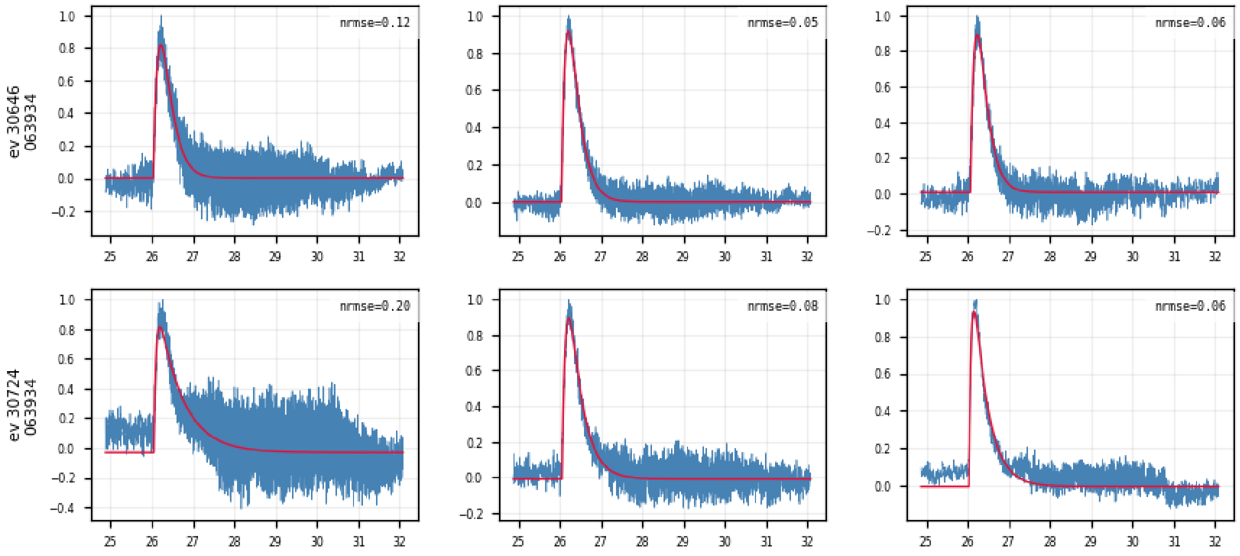
All 13 detectors, PTOFamps spectra (Prof. Saab's study)

Fit quality at a glance — event × channel grids

- One row per event, one column per channel: the low-passed trace (blue) with the fitted pulse model (red) overlaid
- A real event fits well in every channel at once; a noise trigger fails in every channel — a clean handle on which events are real



Noise triggers (Z7): the fit fails in every channel at once



K-line events (Z7): consistent good fits across channels

Per-trace algorithm: low-pass → fit → align

1. **Low-pass** — 100 kHz Butterworth
2. **Normalize** — subtract the baseline, scale the trace to unit peak
3. **Fit** — two-exponential model, all 5 parameters free (including the pretrigger t_0):

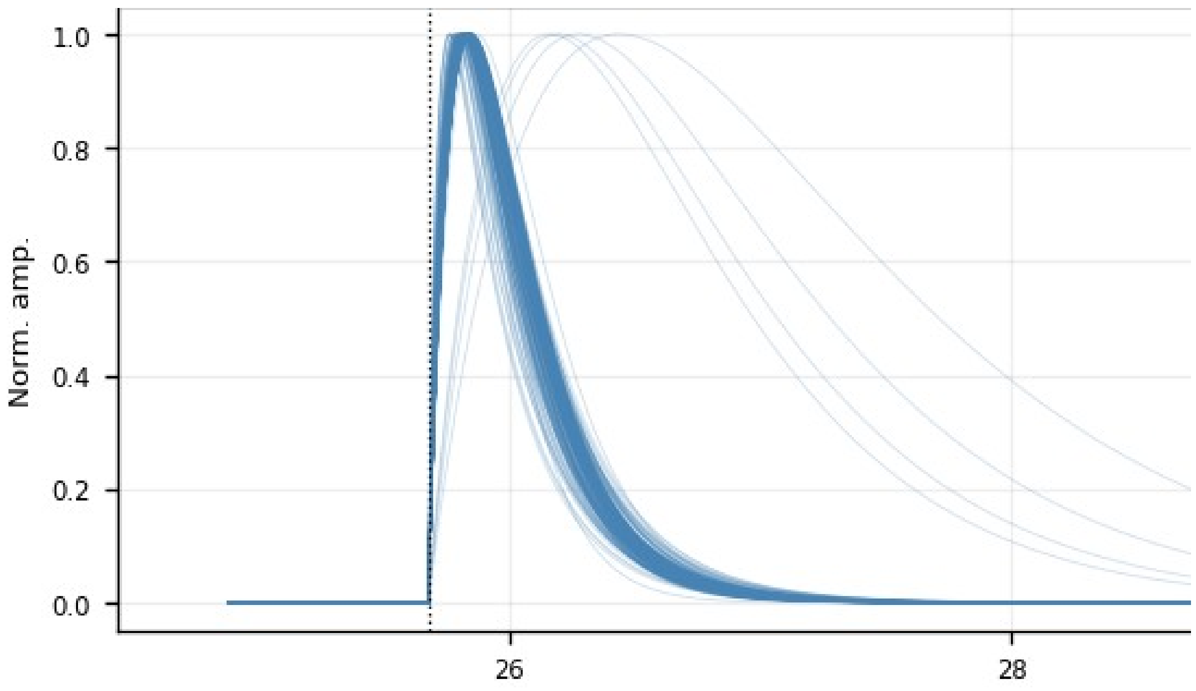
$$y(t) = A\left[\, e^{-(t - t_0)/\tau_{\mathrm{fall}}} - e^{-(t - t_0)/\tau_{\mathrm{rise}}} \,\right] + b$$

t_0 = pretrigger, free within 16050 ± 3000 samples (τ_{rise}, τ_{fall}: rise / fall time)

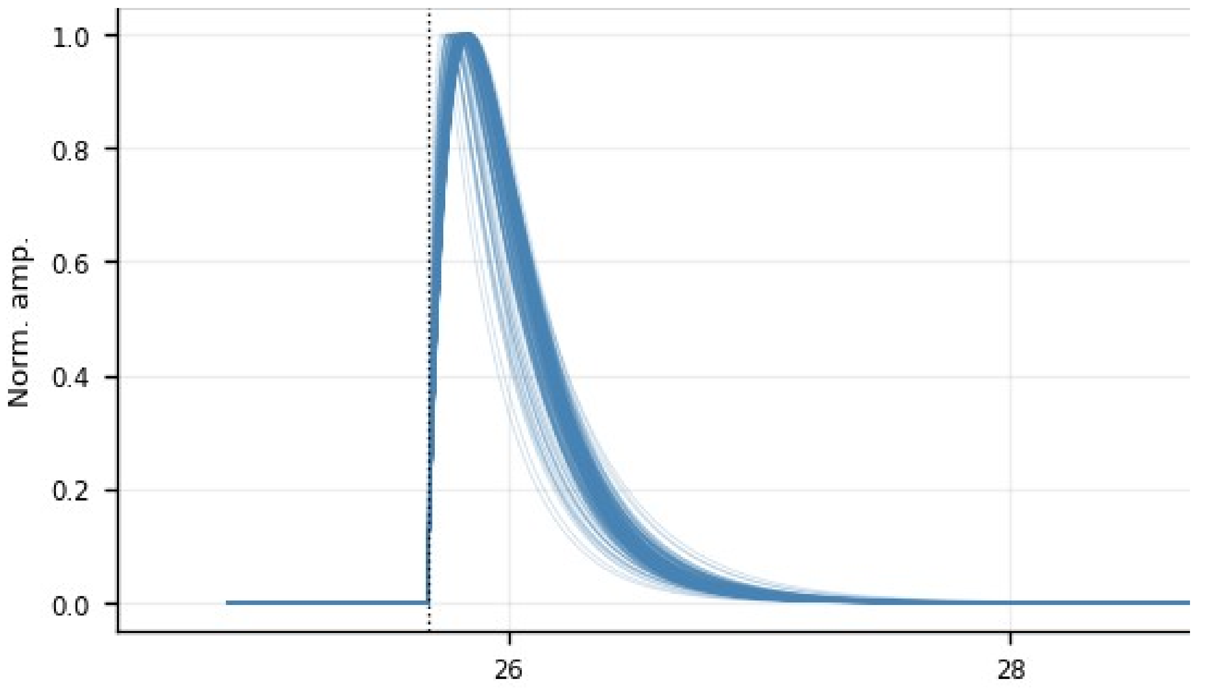
4. **Two quality checks per trace** (recorded here, not cut yet):
   - **fit_ok** — amplitude is positive, and the pulse rises faster than it falls
   - **NRMSE** — RMS of the fit residual ÷ pulse peak (smaller = better fit)

5. Align — shift each trace to a common pretrigger

- Shift the measured trace by (fitted pretrigger − 16050) — a pure translation. Drawn at the common pretrigger, the fitted curves stack into one shape family:



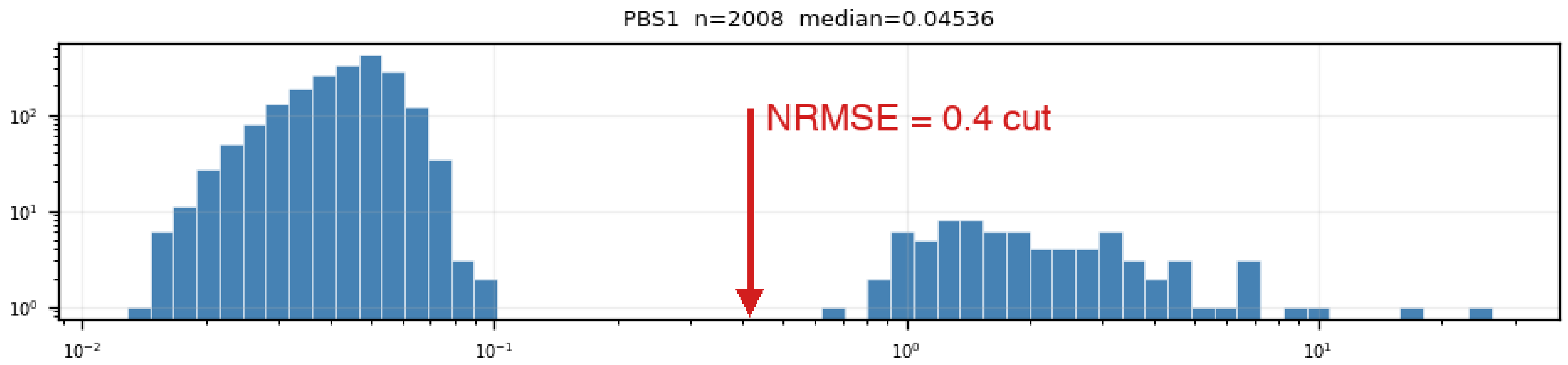
All fit_ok fitted curves (Z7 PBS1) — no cut



After NRMSE ≤ 0.4 (Z7 PBS1) — one tight shape family

Quality cut 1: NRMSE ≤ 0.4 — where the number comes from

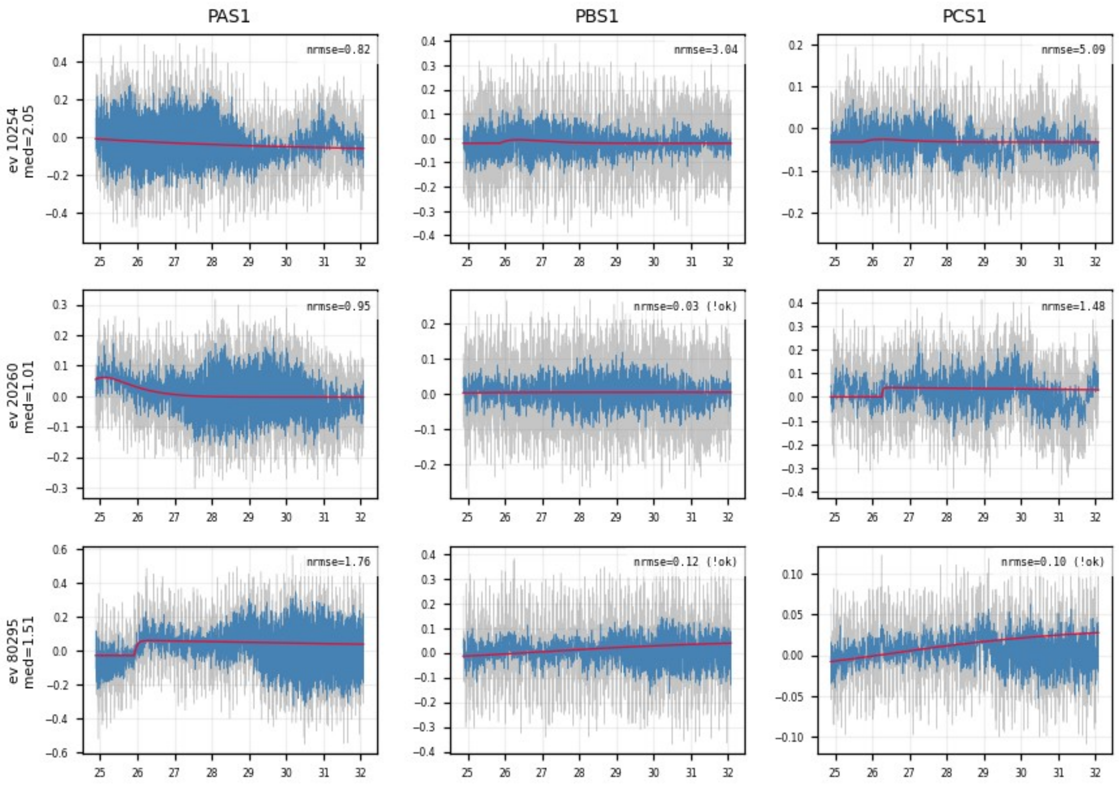
- NRMSE of fit_ok events is bimodal: good fits (median ≈ 0.05–0.1) vs noise triggers (≈ 1–2), valley at ≈ 0.4–0.5
- The cut sits in the valley — it is read off the distribution, not tuned on the templates
- Weak detectors (Z1, Z4, Z6, Z18, Z19, Z22, Z24): same bimodal picture, noise peak dominates



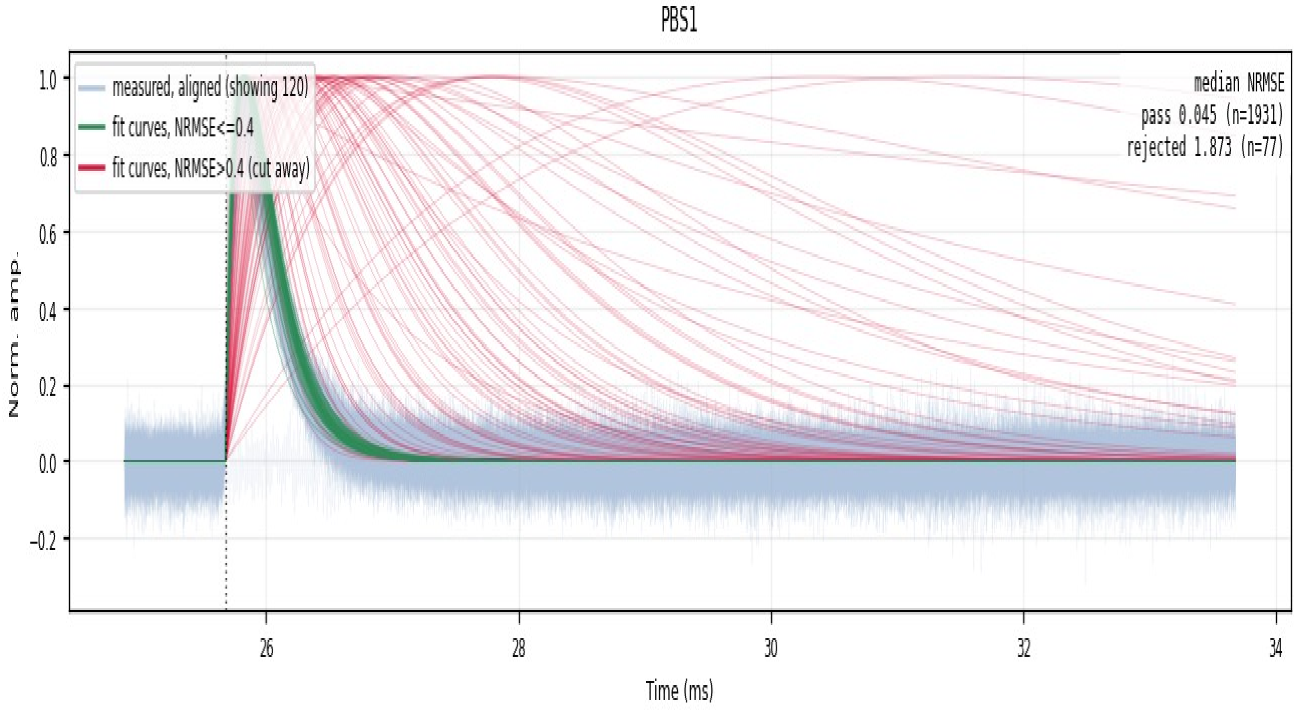
Z7 PBS1: two clean populations, log-log axes

The rejected population is noise — verified in the raw traces

• Event grids of NRMSE-rejected events (median NRMSE > 0.4 across channels): the raw traces show no pulse — the cut removes noise triggers, not physics



Z7: NRMSE-rejected events — raw traces are noise

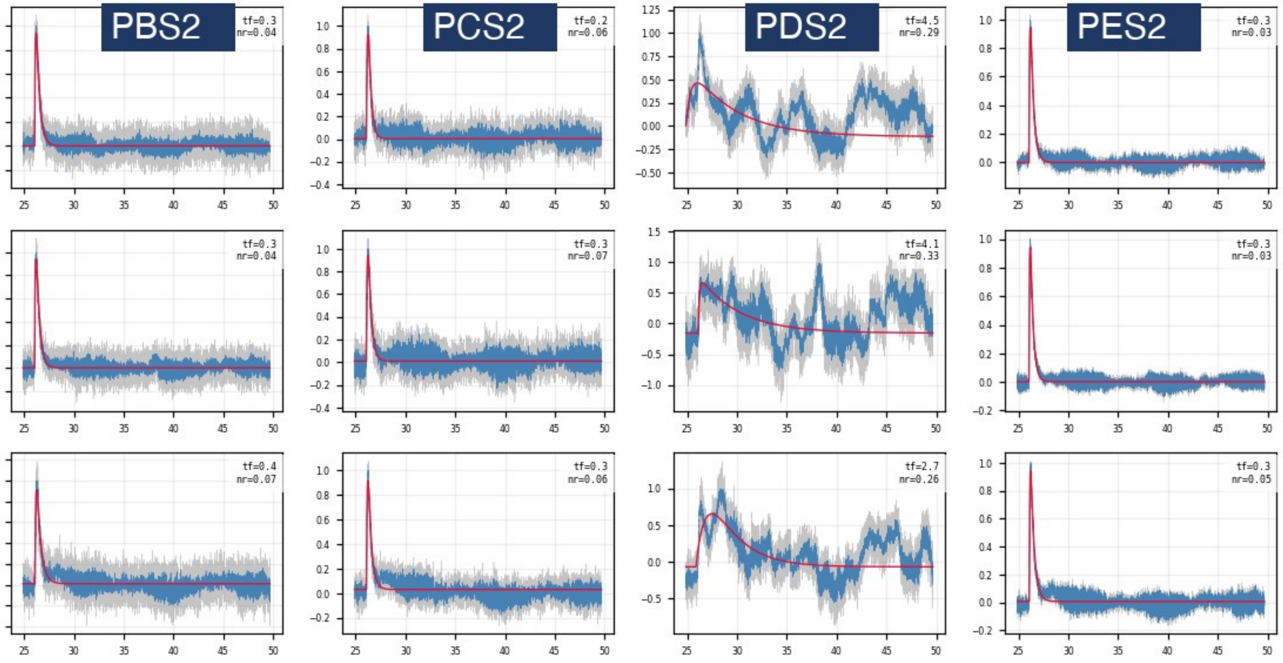


Z7 PBS1 —— **passes the cut (kept)** —— **cut away = rejected**

faint blue = measured traces 1931 kept / 77 cut (~3.8%)

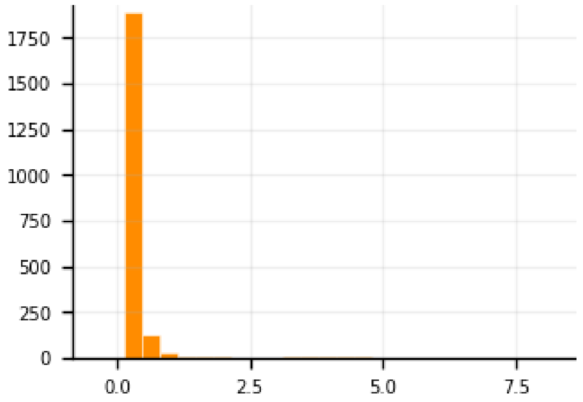
Follow-up 1 — the slow-fall tail is a one-channel artifact

- The post-cut fan still shows slow-fall tails → sample them: NRMSE ≤ 0.4 AND τ_fall > 1.5 ms, 10 random events, raw vs fit drawn in all 12 channels
- The sampled events are real pulses → kept. Their extreme fall times trace to one channel: only PDS2 swings (τ_fall median 0.51 ms vs ≈ 0.25 ms elsewhere) — a low-frequency disturbance
- Conclusion: no τ_fall cut — slow-fall events passing NRMSE stay in; the extreme tail is a one-channel artifact, not a slow pulse

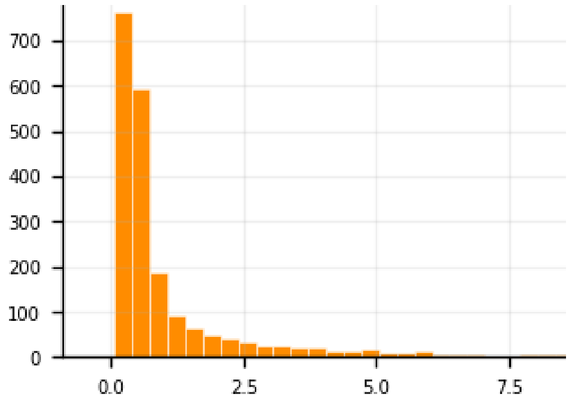


Z7, 3 sampled slow-fall events: normal in PBS2/PCS2/PES2, swings only in PDS2

Fitted τ_fall distribution, same 0–7 ms axis:



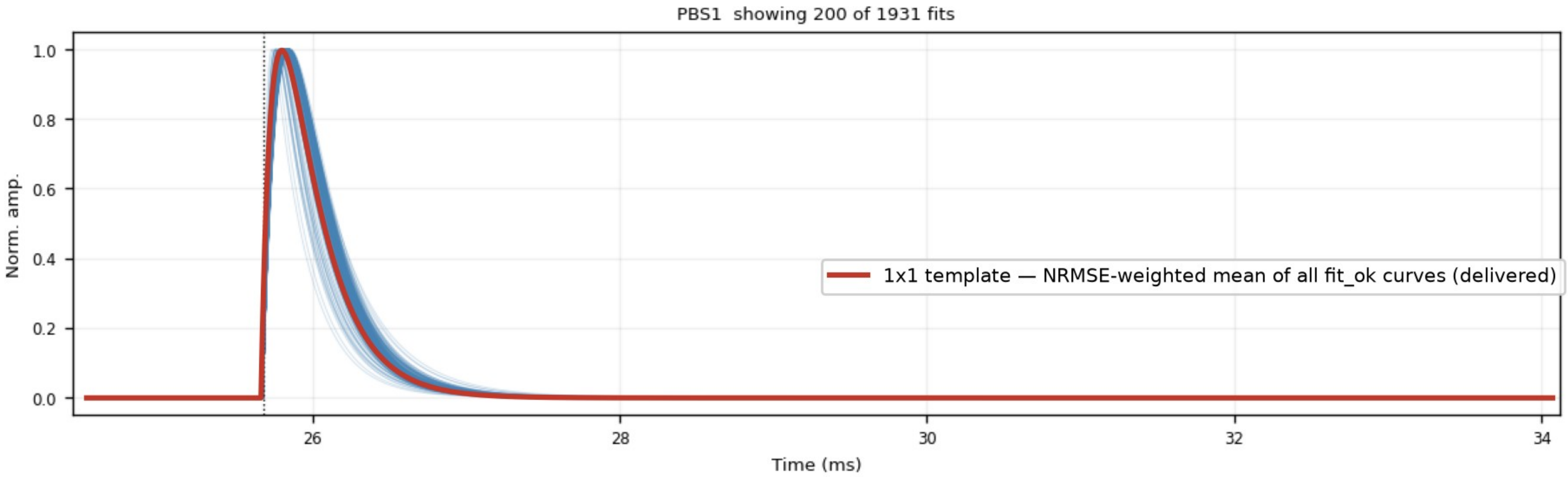
Z7 PAS1 — normal channel: narrow (median 0.25 ms)



Z7 PDS2 — bad channel: broad tail (median 0.51 ms)

Template family 1 — analytic 2-exp, NRMSE-weighted (1x1)

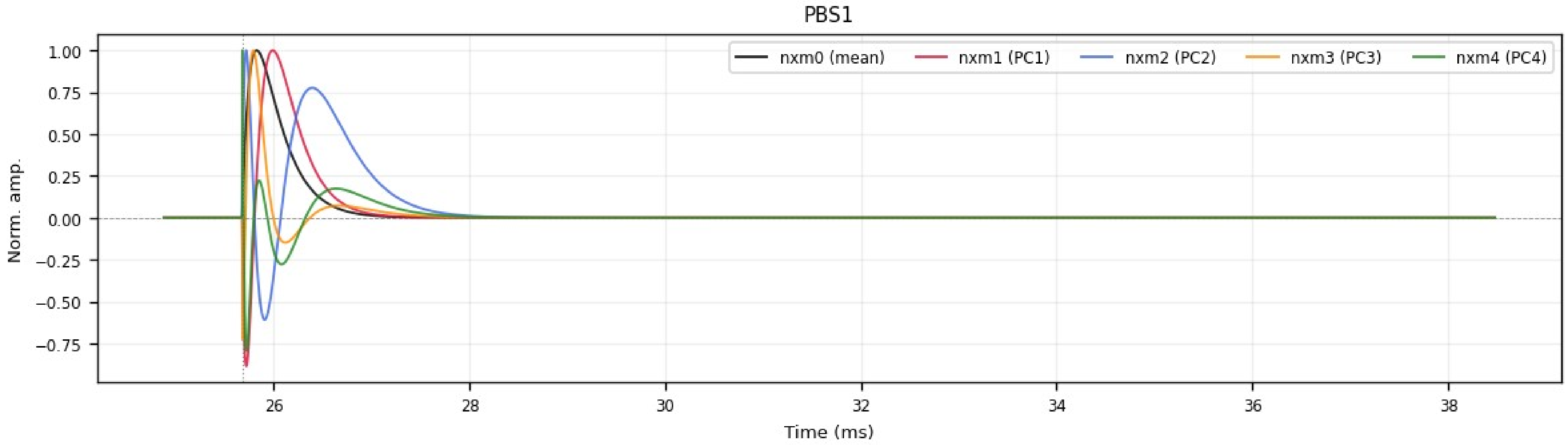
- NRMSE-weighted mean of the fit_ok 2-exp curves
- Smooth & noise-free by construction
- ROOT histograms, peak-normalized (+ summed PT / PS1 / PS2)



Z7 PBS1 — blue: the fitted curves (200 of 1931 drawn); red: the delivered 1x1 template, the NRMSE-weighted mean of all fit_ok curves, read back from the ROOT file

Template family 2 — NxM PCA templates

- Per-channel PCA over the clean fitted curves (fit_ok + NRMSE ≤ 0.4 + τ_rise ≤ 0.3 ms)
- nxm0 = mean shape; nxm1–4 = principal components; real pulse = Σ_i $amp_i \cdot nxm_i$
- PC1 + PC2 ≈ 96–98% of the shape variance; delivered peak-normalized



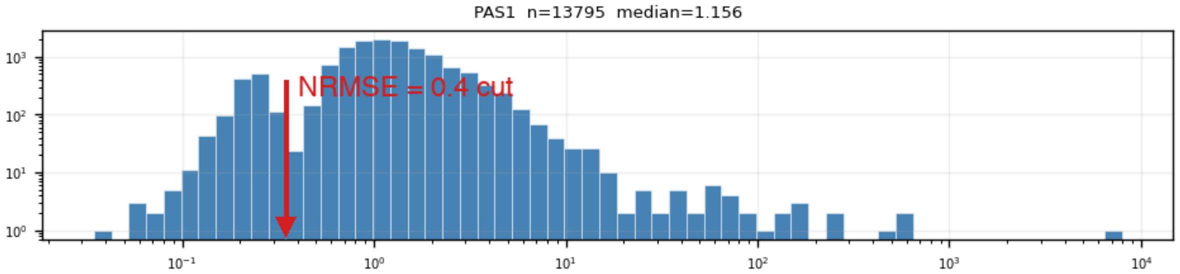
Z7 PBS1: nxm0 (mean) + nxm1–4 (PCs) — the oscillating components encode rise/fall-time variation

Delivered — and what remains

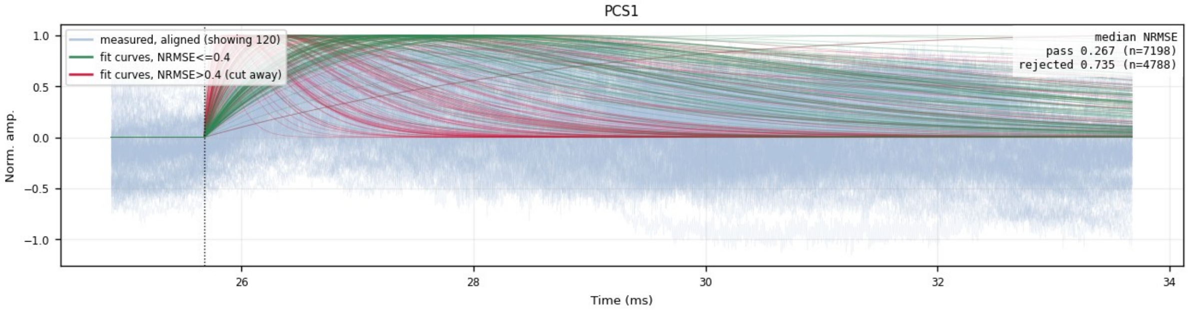
- Two template families built for all 13 detectors: 1x1 (2-exp weighted) + NxM PCA
- Delivered in the official cdmsbats PulseTemplates format
- Cuts read off the data, verified in raw traces: NRMSE ≤ 0.4, $\tau_{rise} \leq 0.3$ ms

Backup — weak detectors (example: Z22)

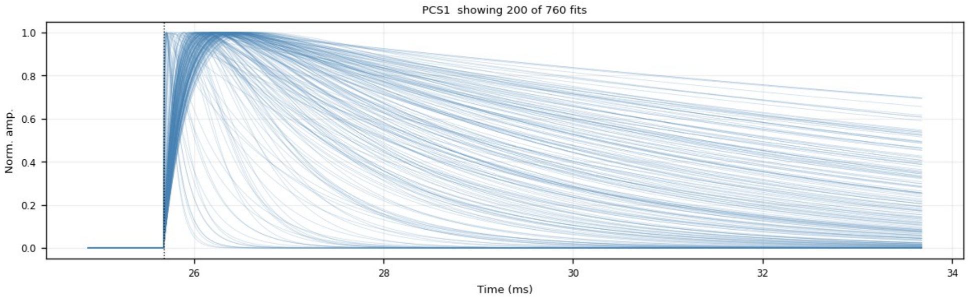
- Z1 / Z4 / Z6 / Z18 / Z19 / Z22 / Z24: K-line sits inside the noise population — the PTOF window admits a noise-dominated mixture
- Same bimodal NRMSE picture, noise peak dominates; the same 0.4 cut digs the real pulses out — Z22 PCS1: 7198 kept / 4788 cut (~40%)
- Kept bundle broader than on quiet detectors; steep red curves = fits latching onto sharp noise spikes, not fast pulses being cut
- τ_rise ≤ 0.3 ms (after NRMSE): removes 55–74% on weak zips (Z22: 71%) vs 1.6% on Z7 — residual slow drift



Z22 PAS1: NRMSE histogram — noise dominates



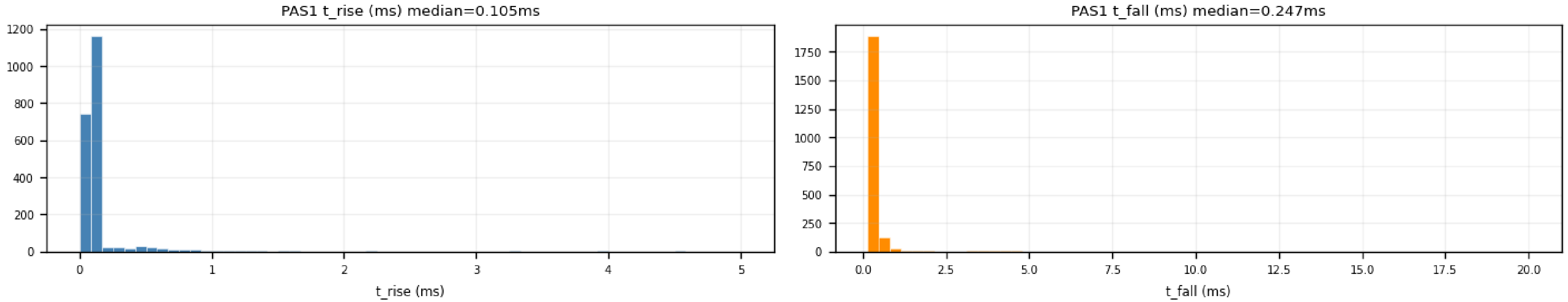
Z22 PCS1: kept (green) vs cut (red)



Z22 PCS1 after NRMSE ≤ 0.4 + τ_rise ≤ 0.3 ms — final shape family (760 fits)

Backup — the τ_rise ≤ 0.3 ms ceiling (PCA input)

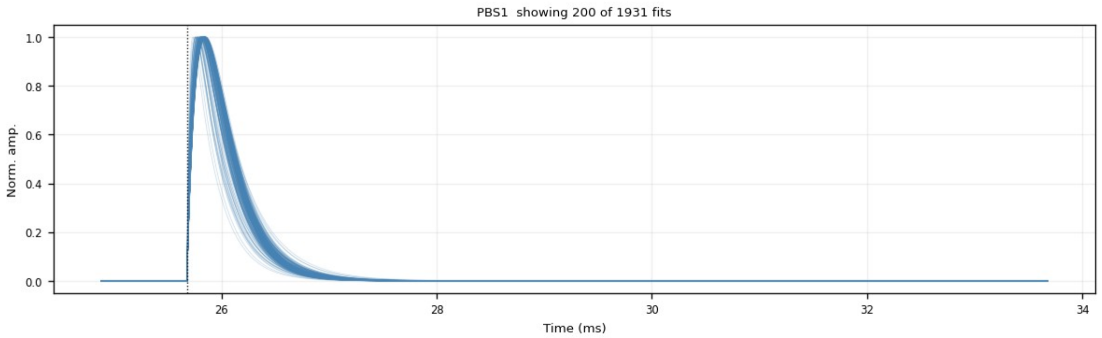
- Slow baseline drift survives NRMSE (a slow 2-exp hugs it) → mimics a slow rise
- Fast pulses at τ_rise ≈ 0.1 ms (p90 ≈ 0.15 ms) → ceiling at 0.3 ms blocks the drift tail
- Removed after the NRMSE cut: Z7 1.6% (quiet 2–6%) — weak 55–74% (Z22 71%) — all zips 54%
- Trade-off: trims the very slowest genuine pulses → decision pending



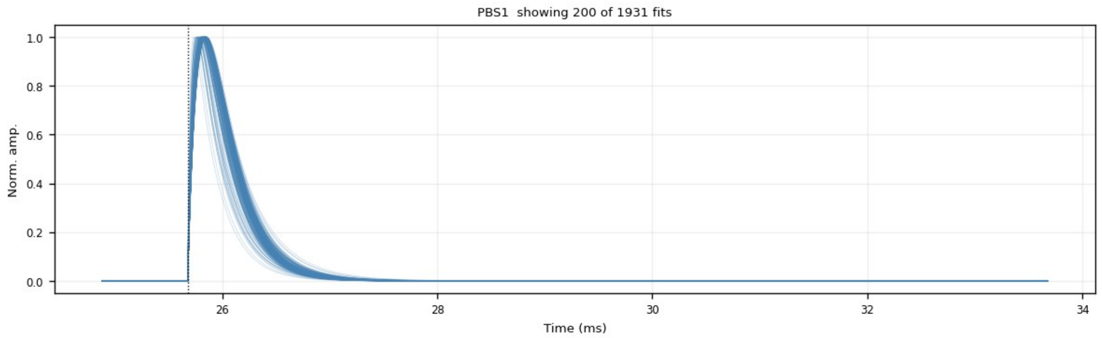
Z7 PAS1: fitted τ_rise (median 0.105 ms) and τ_fall distributions

Backup — τ_rise cut: a good vs a bad channel (Z7)

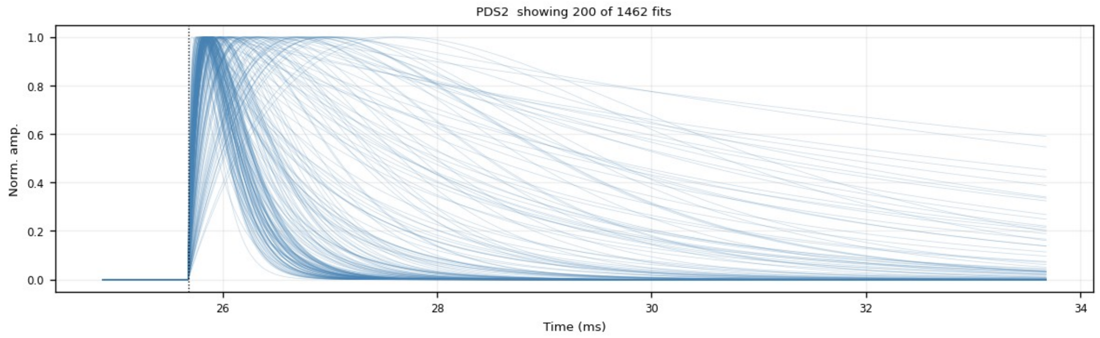
- Good channel PBS1: tight clean bundle — the τ_rise ≤ 0.3 ms cut removes ~0% (nearly free on quiet channels)
- Bad channel PDS2: broad, messy fan — the channel itself is bad; the cut trims the slow-rise curves, 1462 → 1154 (21%)



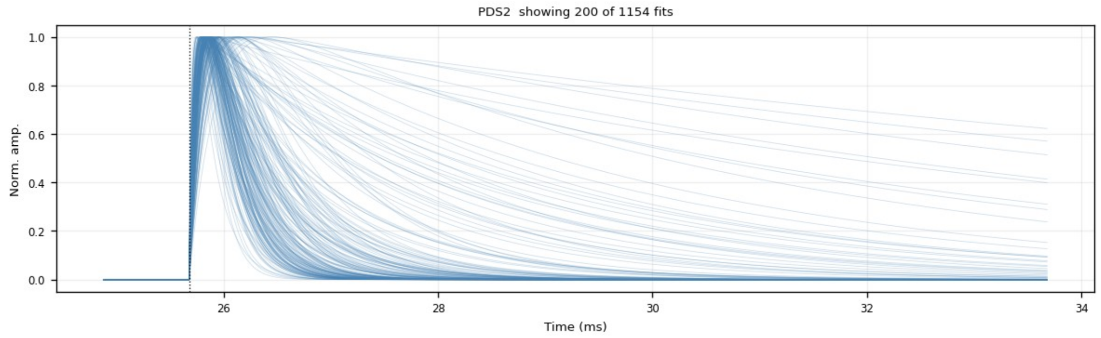
PBS1 (good) — NRMSE ≤ 0.4



PBS1 (good) — + τ_rise ≤ 0.3 ms (unchanged)



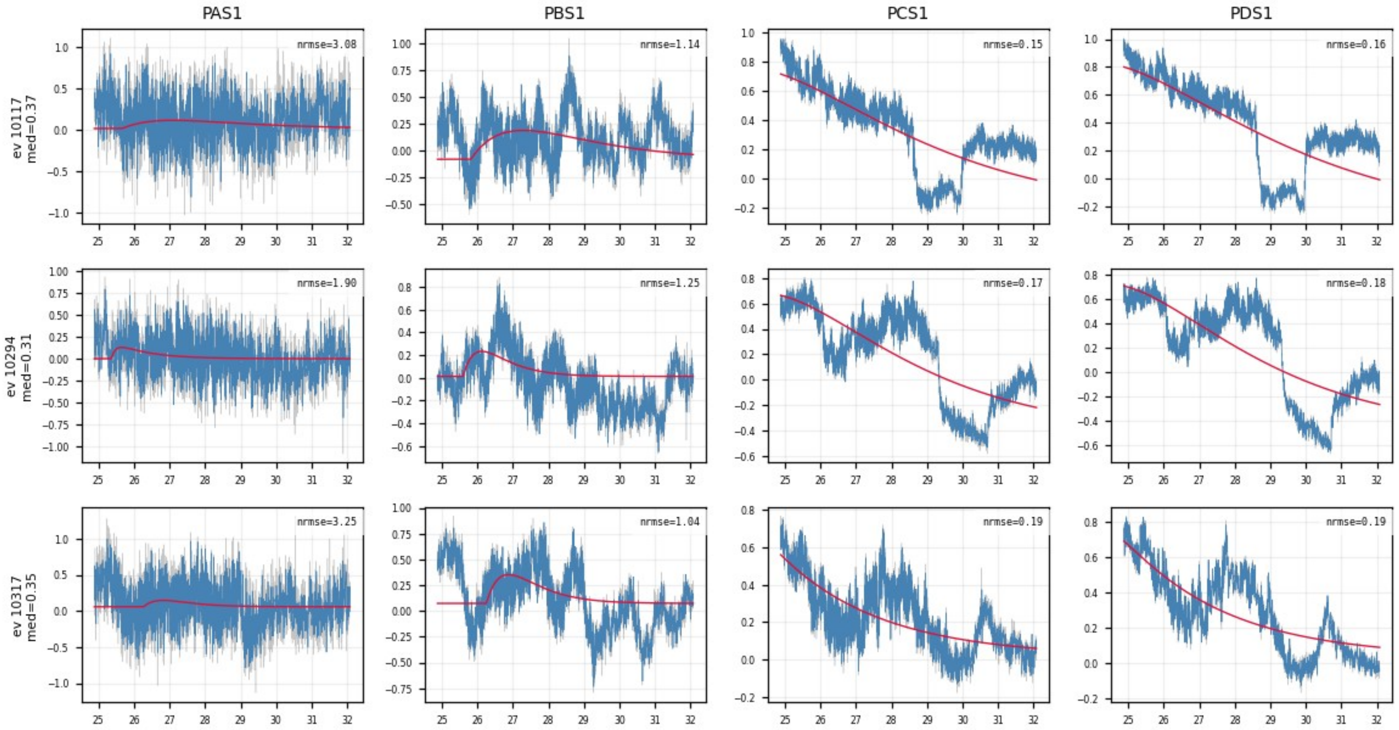
PDS2 (bad) — NRMSE ≤ 0.4



PDS2 (bad) — + τ_rise ≤ 0.3 ms (slow risers trimmed)

Backup — what the τ_rise cut removes: noise NRMSE missed (Z22)

- Exactly the events that PASS NRMSE ≤ 0.4 but are REMOVED by τ_rise ≤ 0.3 ms — the noise 0.4 let through, 0.3 catches
- PCS1 / PDS1 pass at NRMSE ≈ 0.15 (a slow 2-exp hugs a slow baseline drift) — but the raw trace has no clean fast pulse
- Trade-off: the same cut also trims genuine slow-rise pulses — documented, decision pending

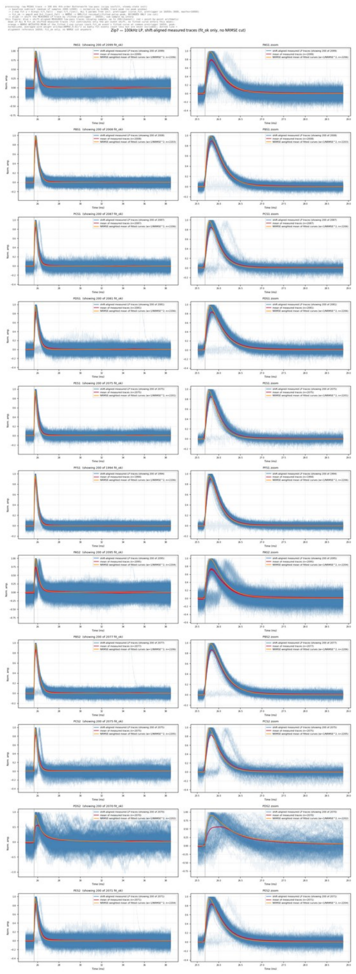


Z22: 3 events × PAS1/PBS1/PCS1/PDS1 — pass NRMSE ≤ 0.4, removed by τ_rise ≤ 0.3 ms; PCS1/PDS1 are pure slow drift

Backup gallery — Z7 · aligned_overlay — shift-aligned measured traces

figure: lp_fit_align results — zip7_lp_aligned_overlay.png

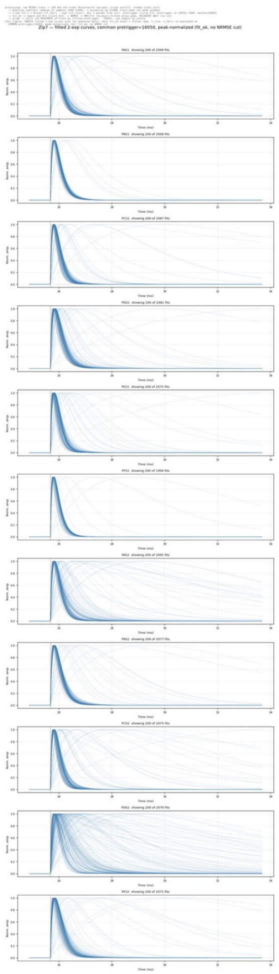
Blue = measured LP traces shifted by (fitted pretrigger − 16050); red = point-by-point mean of ALL fit_ok events; orange = NRMSE-weighted mean of the fitted curves (w = 1/max(NRMSE,0.01)²). Rise edges line up at 16050.



Backup gallery — Z7 · fitted_curves_overlay — fan of fitted 2-exp curves (no cut)

figure: lp_fit_align results — zip7_fitted_curves_overlay.png

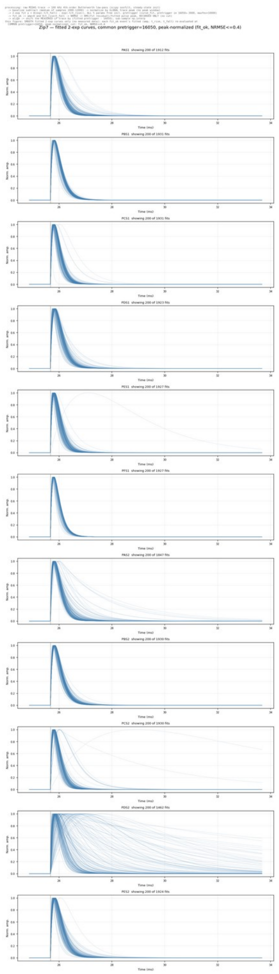
Every fit_ok event's fitted curve re-evaluated at common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Backup gallery — Z7 · fitted_curves_overlay — after NRMSE ≤ 0.4

figure: lp_fit_align results — zip7_fitted_curves_overlay_nrmse0.4.png

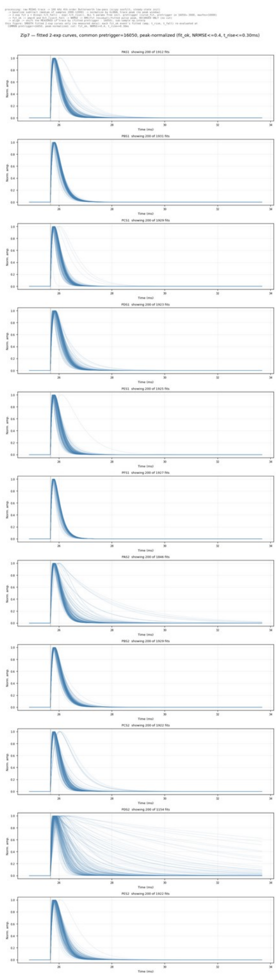
Same fan after the NRMSE cut: the noise fan collapses; the fast-pulse bundle remains.



Backup gallery — Z7 · fitted_curves_overlay — after NRMSE ≤ 0.4 and τ_rise ≤ 0.3 ms

The exact PCA input population: clean fast-pulse bundle only.

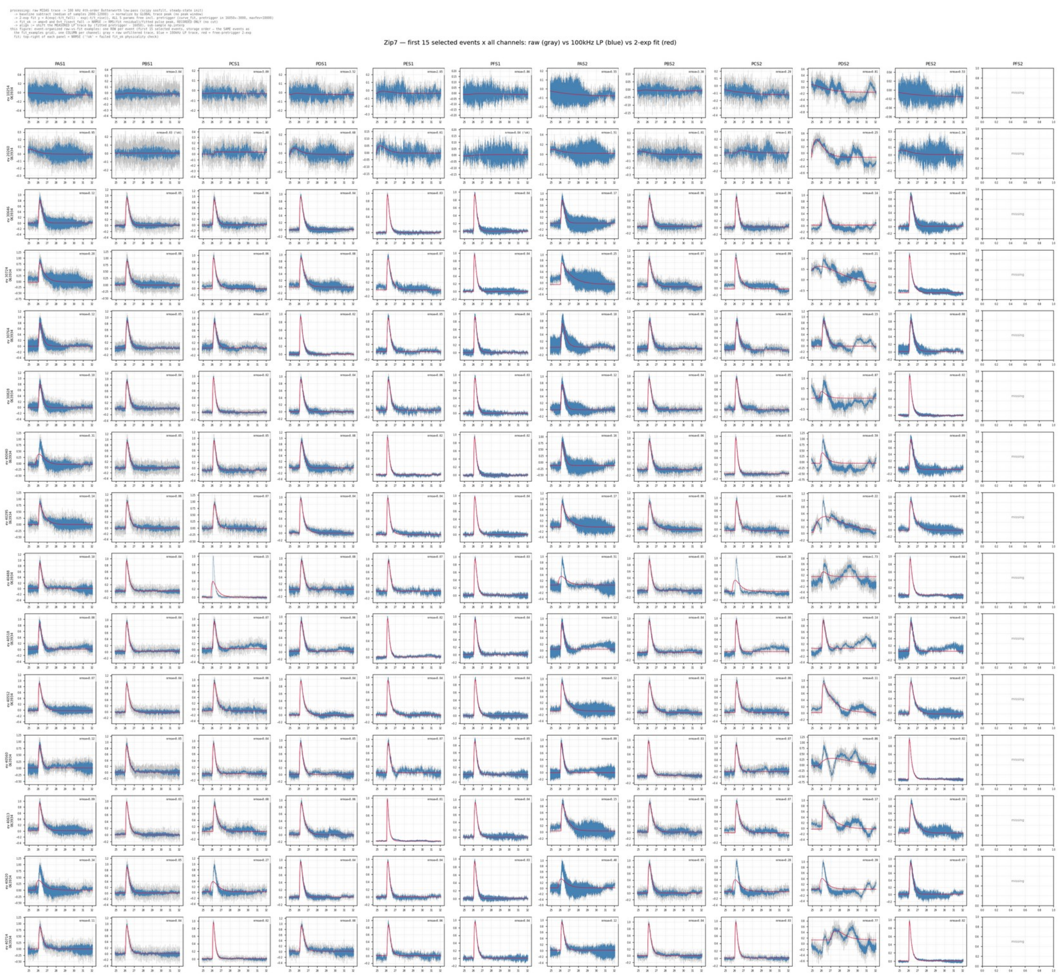
figure: lp_fit_align results — zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png



Backup gallery — Z7 · raw_vs_fit_examples — first 15 events × all channels

Gray = raw unfiltered, blue = 100 kHz LP, red = 2-exp fit; per-panel NRMSE top-right. Same 15 events as fit_examples; noise vs pulse size is directly visible.

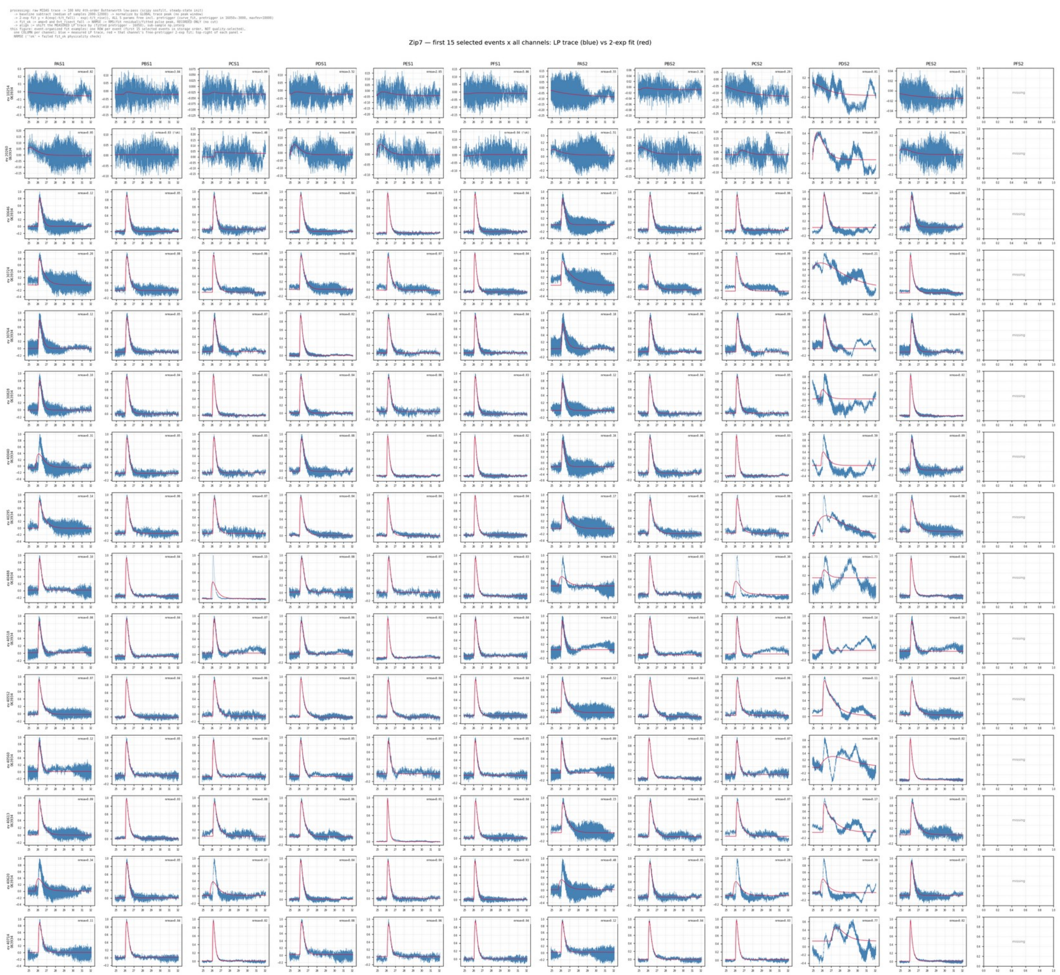
figure: lp_fit_align_results — zip7_raw_vs_fit_examples.png



Backup gallery — Z7 · fit_examples — first 15 events × all channels (LP vs fit)

One ROW per event, one COLUMN per channel. A real event shows a clean pulse and a good fit in every channel simultaneously; a noise trigger fails across the whole row.

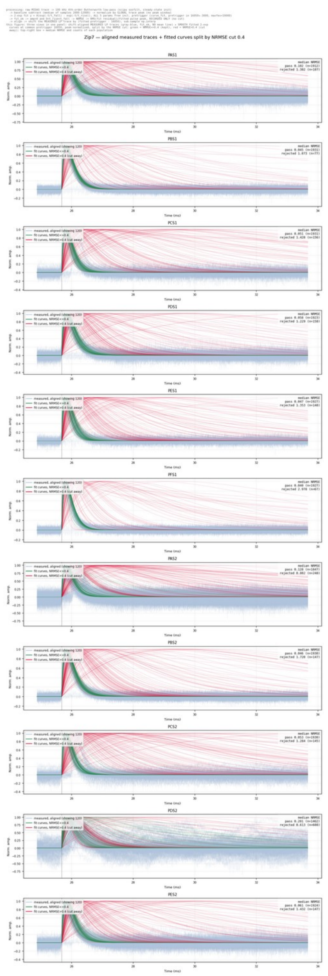
figure: lp_fit_align/results — zip7_fit_examples.png



Backup gallery — Z7 · overlay_fan_cut — measured traces + kept/rejected fits

figure: lp_fit_align results — zip7_overlay_fan_cut_nrmse0.4.png

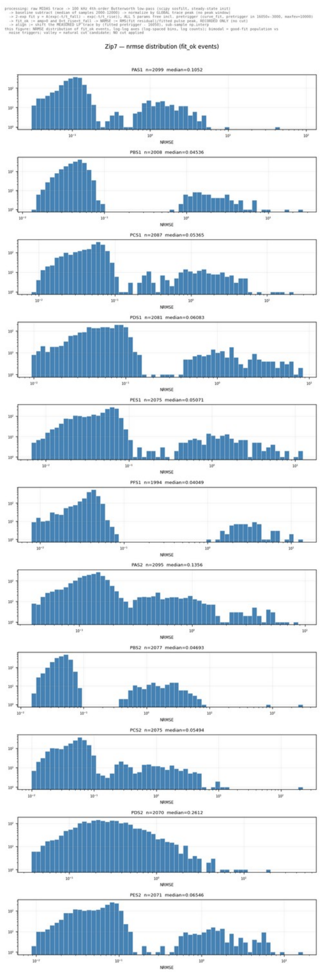
Gray-blue = aligned measured traces; green = fitted curves with NRMSE ≤ 0.4 (kept); red = NRMSE > 0.4 (cut away). Median NRMSE and counts of both populations stamped per channel.



Backup gallery — Z7 · nrmse — NRMSE distribution (log-log)

figure: lp_fit_align results — zip7_nrmse.png

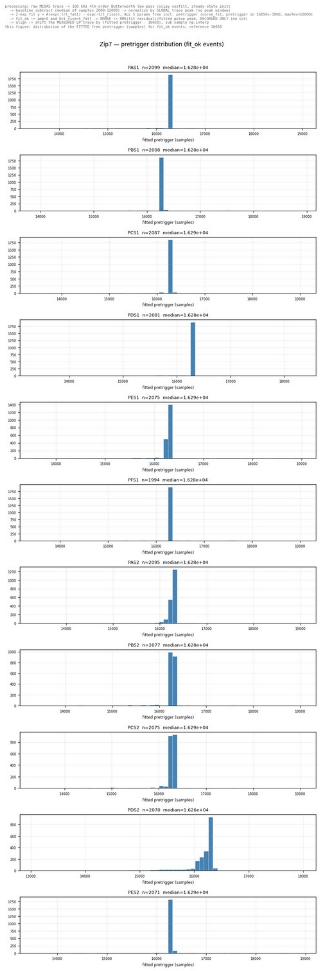
NRMSE = RMS(fit residual)/fitted peak. Bimodal: good-fit population (~0.05–0.1) vs noise triggers (~1–2); the valley at ≈0.4 sets the cut.



Backup gallery — Z7 · pretrigger — fitted onset distribution

figure: lp_fit_align results — zip7_pretrigger.png

The onset is a FREE fit parameter; fitted values cluster ≈230 samples after the nominal 16050, which is why pinning it inside the fit was wrong.



Backup gallery — Z7 · time_constants — fitted τ_rise / τ_fall distributions

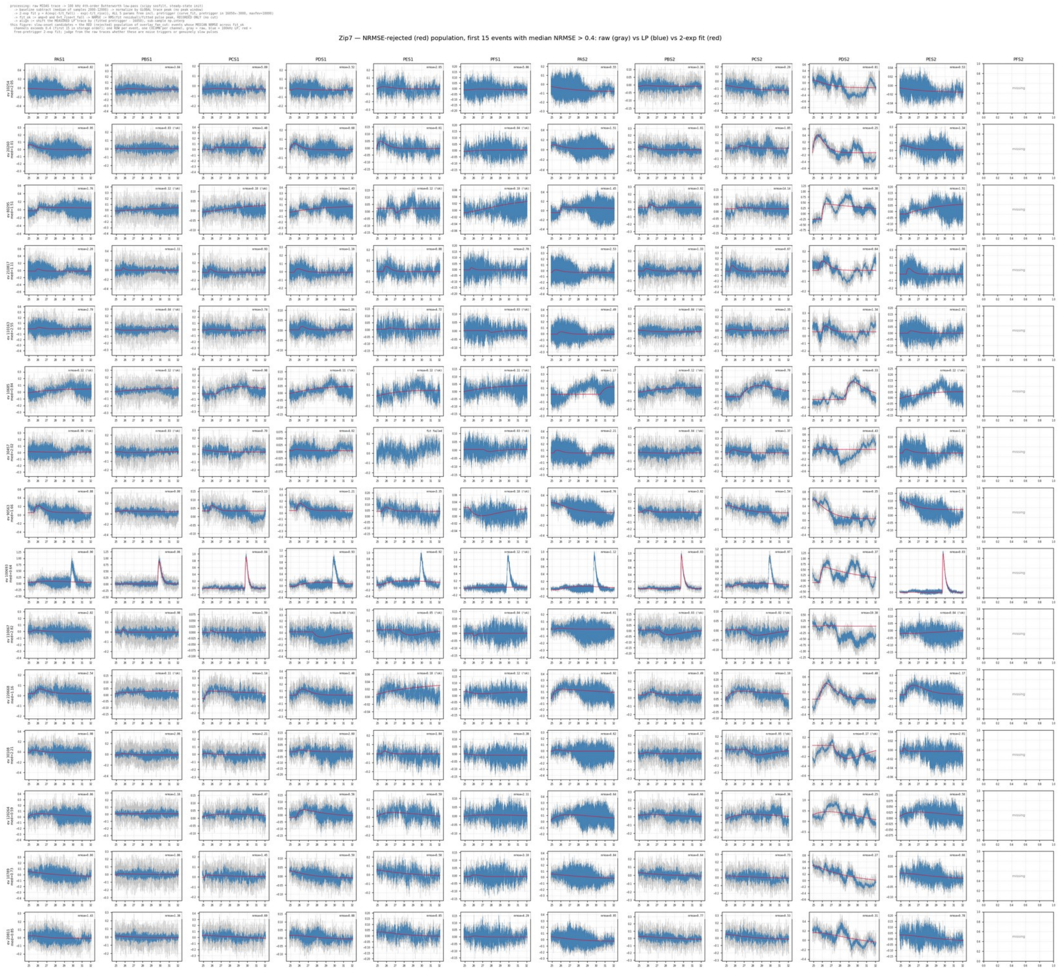
Fast-pulse population sits at τ_rise ≈ 0.1 ms; isolated spikes at the parameter bounds (5 ms / 20 ms) are fits pinned at the fit limits (noise), removed by the NRMSE cut.

figure: lp_fit_align results — zip7_time_constants.png

Backup gallery — Z7 · slow_rise_events — the NRMSE-rejected population

Events whose median NRMSE across channels > 0.4. Raw traces show no pulse in any channel: the rejected population is noise triggers, not physics.

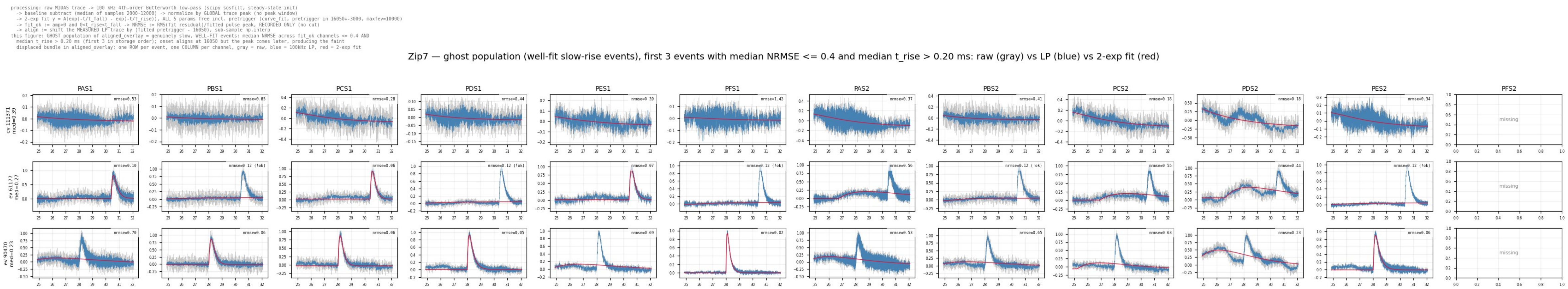
figure: lp_fit_align results — zip7_slow_rise_events.png



Backup gallery — Z7 · shadow_events — well-fit slow-rise (echo-trigger) events

figure: lp_fit_align results — zip7_shadow_events.png

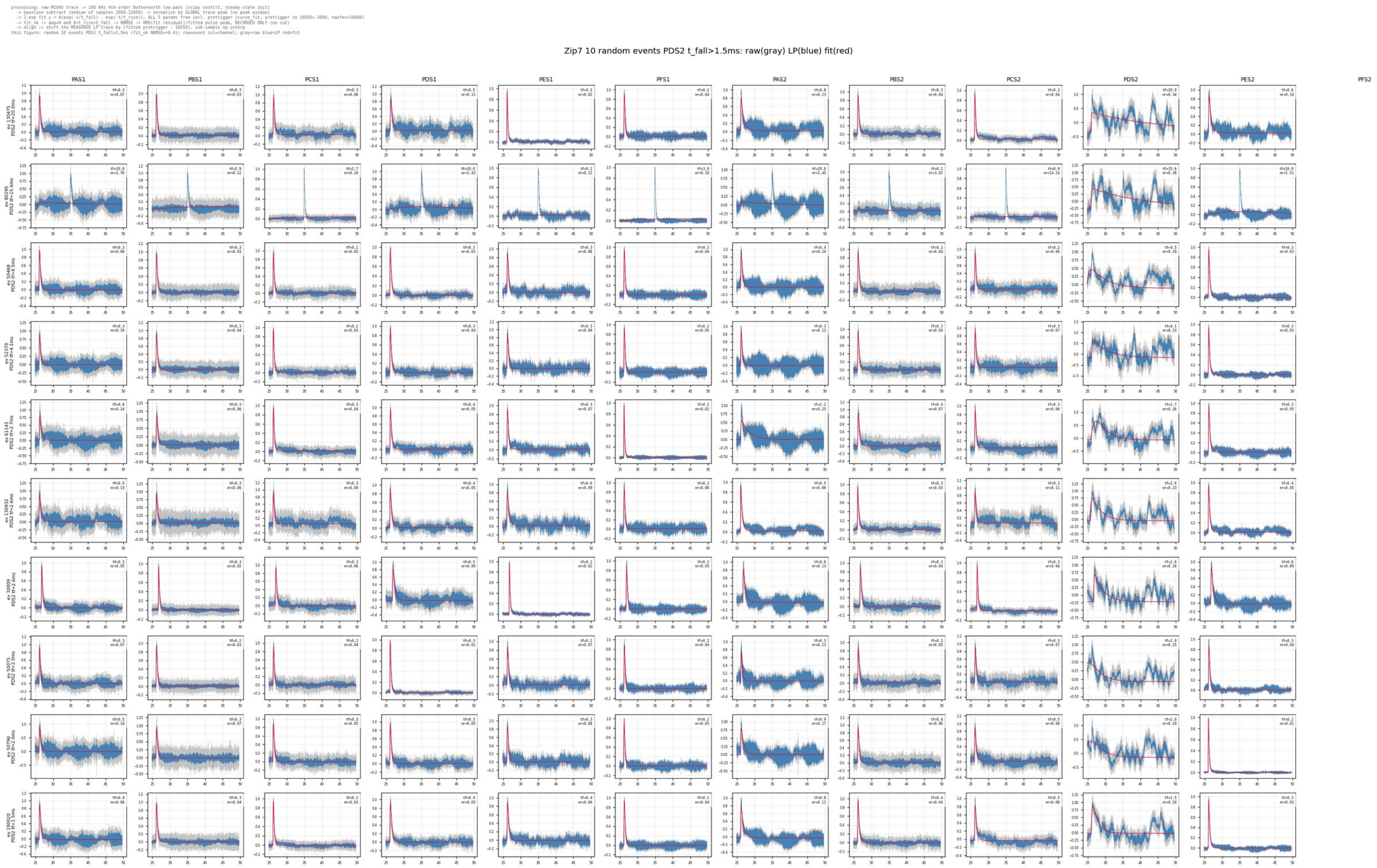
Median NRMSE ≤ 0.4 AND median τ_rise > 0.2 ms: genuinely slow, well-fit pulses — the faint displaced bundle seen in aligned overlays; kept, and exactly what NxM multi-templates are for.



Backup gallery — Z7 · slow_fall_events — long-τ_fall study (PDS2 reference)

figure: lp_fit_align results — zip7_PDS2_slow_fall.png

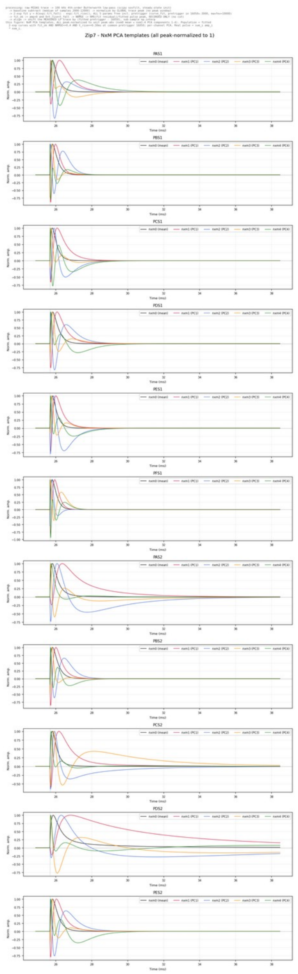
10 random events with PDS2 τ_fall > 1.5 ms drawn in every channel: 11 channels show normal fast pulses; the long fall is a PDS2-only low-frequency artifact — not slow physics, no event-level cut.



Backup gallery — Z7 · deliverables/nxm — PCA templates nxm0–4

figure: lp_fit_align results — zip7_pca_templates.png

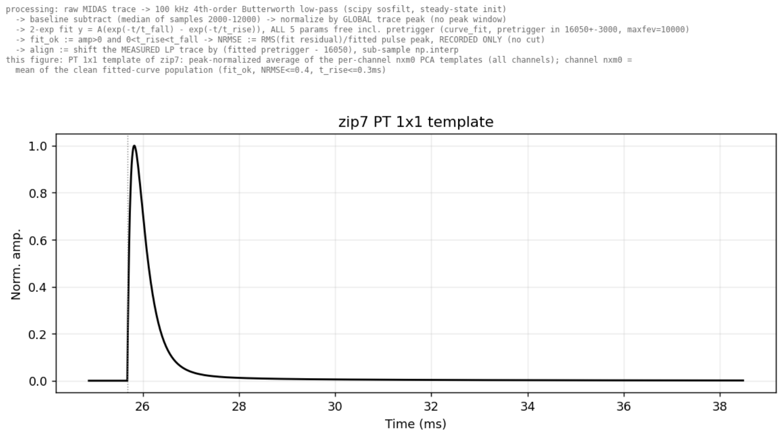
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized to 1; PC1+PC2 capture 96–98% of the shape variance.



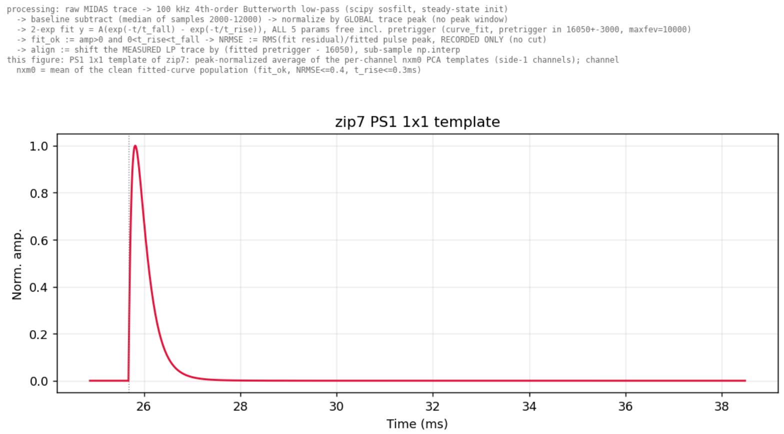
Backup gallery — Z7 · deliverables/1x1 — summed templates PT / PS1 / PS2

Single-template (1x1) summed curves; peak-normalized average of the per-channel nxm0 templates — PT = all channels, PS1 / PS2 = side-1 / side-2 only.

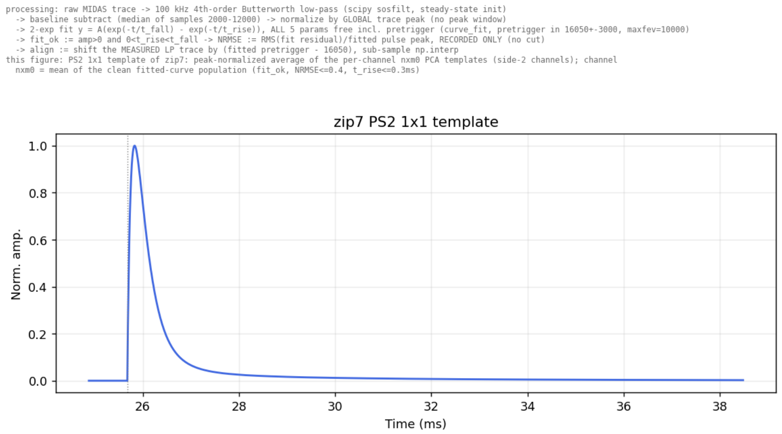
figures: deliverables/1x1/plots/{PT,PS1,PS2}/zip7_*.png



Z7 PT



Z7 PS1



Z7 PS2